

P1.

$$c = p \cdot a + b \pmod{73}$$

Equations.

$$\textcircled{1} \quad 52a + b = 66 \pmod{73}$$

$$\textcircled{2} \quad 20a + b = 15 \pmod{73}$$

$$\textcircled{3} \quad 4a + b = 26 \pmod{73}$$

$$\Rightarrow \textcircled{1} - \textcircled{2}$$

$$\Rightarrow \begin{array}{r} 52a + b = 66 \pmod{73} \\ - 20a - b = 15 \pmod{73} \\ \hline 32a = 51 \pmod{73} \end{array}$$

Mod inverse = 16

$$16 \times 32a = 16 \times 51 \pmod{73}$$

$$a = 816 \pmod{73}$$

OR

$$a = 13 \pmod{73}$$

Substituting $a = 13$ in $\textcircled{3}$

$$\Rightarrow 4(13) + b = 26 \pmod{73}$$

$$b = 26 - 52 \pmod{73}$$

$$b = -26 \pmod{73}$$

OR

$$b = 47 \pmod{73}$$

Thus, $a = 13$; $b = 47$

P4. $M = 467$

$$R_2 = 339, R_3 = 156, R_4 = 390, R_5 = 420, R_6 = 107$$

$\Rightarrow$ Equations.

$$\textcircled{1} \quad 156a + 339b + c = 390 \pmod{467}$$

$$\textcircled{2} \quad 390a + 156b + c = 420 \pmod{467}$$

$$\textcircled{3} \quad 420a + 390b + c = 107 \pmod{467}$$

Now, $\textcircled{2} - \textcircled{1}$

$$\Rightarrow \begin{array}{r} 390a + 156b + c = 420 \pmod{467} \\ - 156a - 339b - c = -390 \pmod{467} \\ \hline 234a - 183b = 30 \pmod{467} \quad \textcircled{A} \end{array}$$

Similarly $\textcircled{3} - \textcircled{2}$

$$\Rightarrow \begin{array}{r} 420a + 390b + c = 107 \pmod{467} \\ - 390a - 156b - c = -420 \pmod{467} \\ \hline 30a + 234b = -313 \pmod{467} \quad \textcircled{B} \end{array}$$

Now $\textcircled{A}$ and $\textcircled{B}$,

$$\textcircled{A} \quad 234a - 183b = 30 \pmod{467}$$

$$\textcircled{B} \quad 30a + 234b = -313 \pmod{467}$$

$$\text{Now, } \frac{234}{30} = \frac{5}{39}$$

We can multiply $\textcircled{A}$ by 5, and $\textcircled{B}$ by 39 to solve for b with elimination.

$$\Rightarrow \begin{array}{r} 1170a - 915b = 150 \pmod{467} \\ 1170a + 9126b = -12207 \pmod{467} \end{array}$$

Subtracting $\textcircled{A}$ from $\textcircled{B}$ yields,

$$10041b = -12357 \pmod{467}$$

Which simplifies to,

$$234b = 252 \pmod{467}$$

Mod inverse = 2,

$$\Rightarrow 234 \times 2b = 252 \times 2 \pmod{467}$$

$$b = 37 \pmod{467}$$

$\Rightarrow$ Substituting $b = 37$ in $\textcircled{B}$

$$\Rightarrow 30a + 234(37) = -313 \pmod{467}$$

$$\Rightarrow 30a + 8658 = -313 \pmod{467}$$

$$\Rightarrow 30a = -313 - 8658 \pmod{467}$$

$$\Rightarrow 30a = -8971 \pmod{467}$$

OR

$$30a = 369 \pmod{467}$$

Mod inverse = 109

$$\Rightarrow 30 \times 109a = 369 \times 109 \pmod{467}$$

$$a = 40221 \pmod{467}$$

$$a = 59 \pmod{467}$$

Now, Substituting $a = 59$ and $b = 37$ in $\textcircled{1}$

$$\Rightarrow 156(59) + 339(37) + c = 390 \pmod{467}$$

$$\Rightarrow 9204 + 12543 + c = 390 \pmod{467}$$

$$\Rightarrow c = 390 - 21747 \pmod{467}$$

$$\Rightarrow c = -21357 \pmod{467}$$

OR

$$c = 125 \pmod{467}$$

Now,

For R_1 ,

$$R_2 = 59R_1 + 37R_1 + 125 \pmod{467}$$

$$\Rightarrow 156 = 59(339) + 37R_1 + 125 \pmod{467}$$

$$\Rightarrow 156 = 37R_1 + 20126 \pmod{467}$$

OR

$$156 = 37R_1 + 45 \pmod{467}$$

$$111 = 37R_1 \pmod{467}$$

$$\Rightarrow R_1 = 3$$

For R_0 ,

$$R_2 = 59R_0 + 37R_0 + 125 \pmod{467}$$

$$\Rightarrow 339 = 59(3) + 37R_0 + 125 \pmod{467}$$

$$\Rightarrow 339 = 302 + 37R_0 \pmod{467}$$

$$\Rightarrow 37 = 37R_0 \pmod{467}$$

$$\Rightarrow R_0 = 1$$

For R_7 ,

$$R_7 = 59R_6 + 37R_5 + 125 \pmod{467}$$

$$= 59(107) + 37(420) + 125 \pmod{467}$$

$$= 6313 + 15540 + 125 \pmod{467}$$

$$= 21978 \pmod{467}$$

OR

$$= 29 \pmod{467}$$

$$R_7 = 29$$

Thus, $a = 59$; $b = 37$; $c = 125$; $R_0 = 1$; $R_1 = 3$

and $R_7 = 29$